Programming Language CPCS301

2nd Semester 2021 Mid Exam Female Dep.

This copy is reassembled by students of CPCS301 female department mid exam to help other students to practice exam for the subject. The mid exam covered the ideas of Ch.1 –Ch.3 – Ch.5

May Allah bless you and Good luck.

SOLVED SHEET

<<Solution Are Indicated in Blue>>

King Abdulaziz University
Faculty of Computing & information Technology
Computer Science Department

Name : ....

Student ID : ....

Section : DAR / AAR / EAR

Total 20 mark

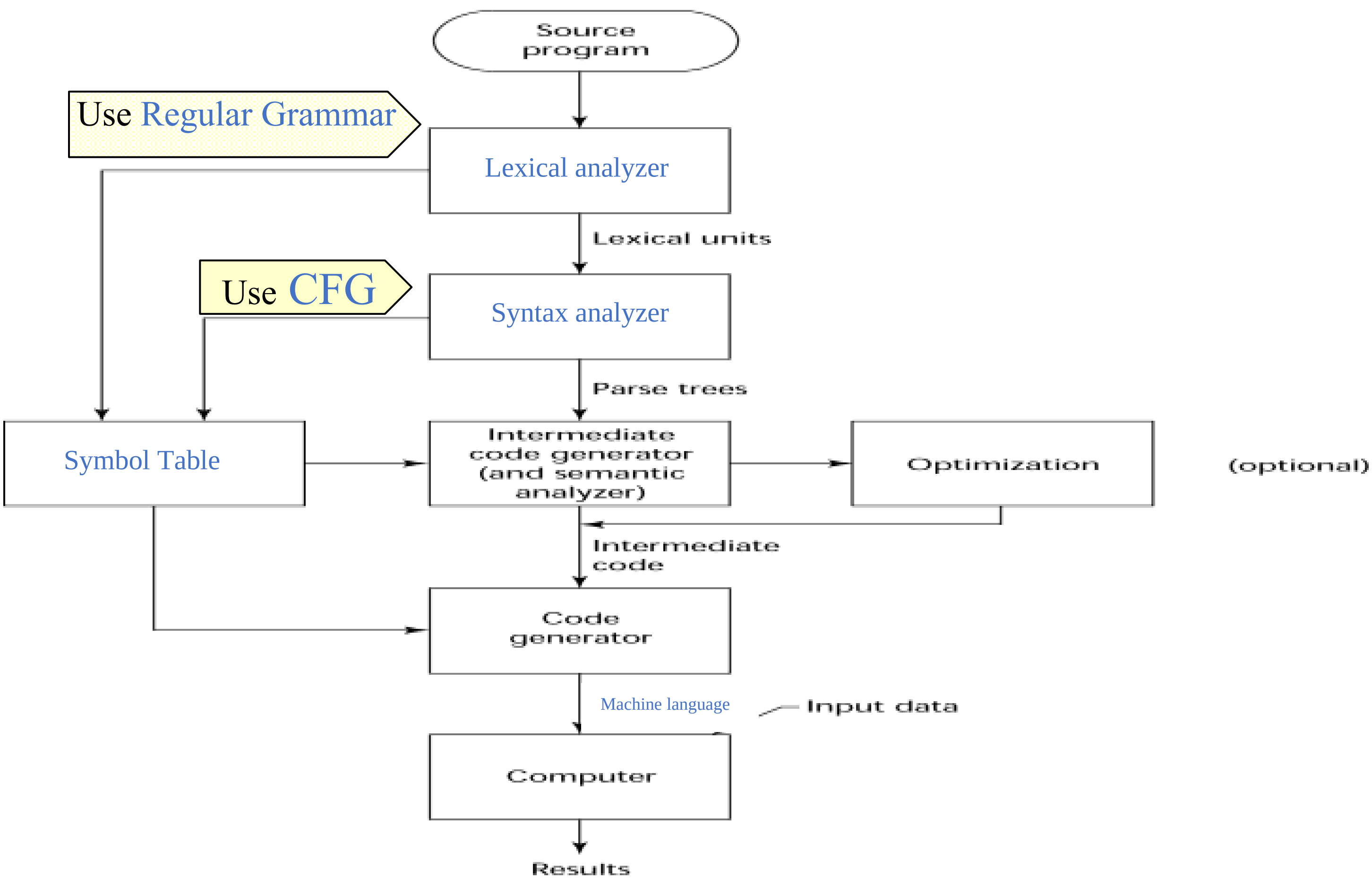
Q1) Write the criteria for evaluating a programming language:

1- Readability 2- Writability 3- Reliability 4- Cost

Q2) Match between the domain and the language

1- Imperative	4	Lisp, Scheme
2- Logic	3	Java, C++
3- Object Oriented	1	C, Pascal
4- Functional	2	Prolog, Algol

Q3) Fill the space in the figure that represent the combinational process:



Q4) Using the grammar :

< identifier> → $< letter> { <letter> | <digit> }
< letter> → a | b | c | d | e
<digit> → 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

determine if the following statements are produced by the previous grammar or not by filling the blanks with (yes / no)

1- (Yes) $a1234
2- (Yes) $abcd1234
3- (No) $$abc123
4-(No) d123
5- (No) $L123
6- (No) $123ab$

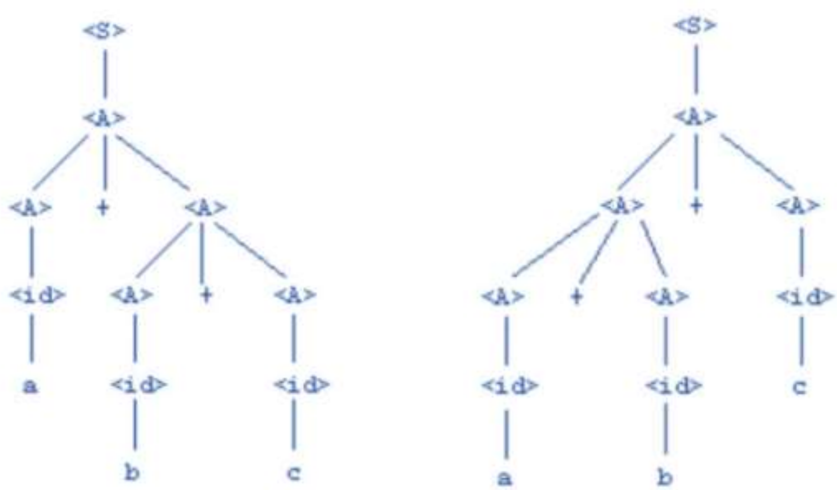
Q5) a. Describe when a grammar is considered an ambiguous grammar?

A grammar is ambiguous if and only if it generates a sentential form that has two or more distinct parse trees.

b. give two reasons that make a grammar ambiguous:
1- Grammar lacks semantic of operator precedence
2- Grammar lacks semantic of associativity

Q6) determine if the following expression is ambiguous or not for the expression a+b+c, show your work:

<E> → <T>
<T> → <T> + <T> | <id>
<id> → a | b | c



Since it produced more than one tree for the same expression, it is ambiguous.

Q7) determine if the following sentence is true or false (True / False):

(T) Binding is the association of between a symbol and operation, or an attribute with entity

(F) In the expression x = y, L-side x represent the value and the R-side y represent the address.

(F) Scoop of dynamic binding inside a function starts when the variable is bound to the memory cell and remain there until the program terminate

(T) reserved words are preferable than keywords since they are more redefine

(F) Pointers are used in implicate heap dynamic binding.

(T) Scoop is range of statements of which a variable is visible.

Q8) This code is written in JavaScript used a static binding, determine each sub-call scoop of variables within Sub1, Sub2 and Sub3, and determine from where are they defined:

```
Main(){
int x,y,z

Sub1(){
int a,b,z;

      Sub2(){
       int b,c,d;


         }
}

Sub3(){
int a,y,z;

   }


}
```

Sub1(): a,b,z from Sub1() and x,y from Main()

Sub2(): b,c,d from Sub2() and a,z from Sub1() and x,y from Main()

Sub3(): a,y,z from Sub3() and x from Main()

Finish, for any comment or if you found a mistake, please contact us, good luck